AMR Ranker - Antibiotic Resistance Report

Research and Educational Use Only - Not for Clinical Decision-Making

Report generated: 2026-06-15 22:50

Patient Information

Patient ID	TEST-001
Age	45
Sex	Male
Weight (kg)	75.0
Pregnant	No
Age Group	Adult
Comorbidities	diabetes
Known Allergies	penicillin
Infection Site	Respiratory

Isolate Information

Species	Streptococcus pneumoniae
Isolation Type	Clinical
Country	USA
Collection Year	2022

Detected Resistance Genes:

gene_mef(a), gene_erm(b), gene_pbp2b

Ranked Antibiotic Recommendations

Rank	Antibiotic	Drug Class	Resist. %	Effect. %	Score	Recommendation
1	Ceftriaxone	Cephalosporin	26.8%	73.2%	0.73	Recommended
2	Tetracycline	Tetracycline	28.0%	72.0%	0.72	Recommended
3	Clindamycin	Lincosamide	30.2%	69.8%	0.70	Use with Caution
4	Tmp_smx	Sulfonamide	34.0%	66.0%	0.66	Use with Caution
5	Erythromycin	Macrolide	42.6%	57.4%	0.57	Use with Caution
6	Meropenem	Carbapenem	52.2%	47.8%	0.48	Use with Caution
7	Amoxicillin	Beta-lactam (Penicillin)	45.5%	54.5%	0.00	Contraindicated (Allergy)
8	Penicillin	Beta-lactam (Penicillin)	38.8%	61.2%	0.00	Contraindicated (Allergy)

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Patient-Specific Ranking Details

#1 Ceftriaxone - Recommended

The model predicts a 26.8% probability of resistance for this isolate, corresponding to a base effectiveness score of 73.2%. No resistance genes specifically associated with ceftriaxone were detected in this isolate; the prediction reflects broader genomic and population-level resistance patterns learned from the training data.

#2 Tetracycline - Recommended

The model predicts a 28.0% probability of resistance for this isolate, corresponding to a base effectiveness score of 72.0%. No resistance genes specifically associated with tetracycline were detected in this isolate; the prediction reflects broader genomic and population-level resistance patterns learned from the training data.

#3 Clindamycin - Use with Caution

The model predicts a 30.2% probability of resistance for this isolate, corresponding to a base effectiveness score of 69.8%. erm(b) (ribosomal methylase) is present, which modifies the bacterial ribosome to block macrolide and lincosamide binding, conferring high-level resistance to both erythromycin and clindamycin (MLSb phenotype).

#4 Tmp_smx - Use with Caution

The model predicts a 34.0% probability of resistance for this isolate, corresponding to a base effectiveness score of 66.0%. No resistance genes specifically associated with tmp_smx were detected in this isolate; the prediction reflects broader genomic and population-level resistance patterns learned from the training data.

#5 Erythromycin - Use with Caution

The model predicts a 42.6% probability of resistance for this isolate, corresponding to a base effectiveness score of 57.4%. mef(a) (macrolide efflux pump) is present, which actively exports macrolide antibiotics out of the bacterial cell and reduces erythromycin effectiveness. erm(b) (ribosomal methylase) is present, which modifies the bacterial ribosome to block macrolide and lincosamide binding, conferring high-level resistance to both erythromycin and clindamycin (MLSb phenotype).

#6 Meropenem - Use with Caution

The model predicts a 52.2% probability of resistance for this isolate, corresponding to a base effectiveness score of 47.8%. No resistance genes specifically associated with meropenem were detected in this isolate; the prediction reflects broader genomic and population-level resistance patterns learned from the training data.

#7 Amoxicillin - Contraindicated (Allergy)

This antibiotic is contraindicated due to a documented penicillin allergy, and is therefore not recommended regardless of its predicted resistance profile.

#8 Penicillin - Contraindicated (Allergy)

This antibiotic is contraindicated due to a documented penicillin allergy, and is therefore not recommended regardless of its predicted resistance profile.

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Disclaimer

This report is generated for research and educational purposes only. It is not a substitute for clinical diagnosis, laboratory antimicrobial susceptibility testing, or professional medical advice. Predictions are based on a research dataset and machine learning models that have not been validated for clinical use. Always consult a qualified healthcare professional and confirmed laboratory AST results before making treatment decisions. No patient data is stored - all inputs are processed in real time and discarded.